



TED UNIVERSITY

CMPE491

Senior Project I

HorusEye with

Project Proposal

Submission Date

31.10.2025

1 Name of the project

HorusEye - AI-Based Exam Proctoring System

2 The URL of the project web page

<https://v0-horus-eye-coming-soon-page.vercel.app/>

3 Names of the team members

- Gizem Nur İpek
- Çağla Abazaoğlu
- Taha Kürşat Öztürk
- Tuğba Hilal Kırer
- Ali Sahil

4 Names of the supervisor and jury members

- Fırat Akba – Supervisor
- Tolga Kurtuluş Çapın– Jury Member
- Eren Ulu– Jury Member

5 Description of the project

Academic integrity is a fundamental pillar of educational institutions, and maintaining fair examination conditions is crucial for ensuring the validity and reliability of assessment outcomes. Traditional exam proctoring methods rely heavily on human supervision, which can be inconsistent, resource-intensive, and prone to human error. As educational institutions increasingly adopt hybrid and online learning models, there is a growing need for automated, reliable, and scalable solutions that can effectively monitor exam sessions while reducing the burden on human proctors.

Our project addresses this critical need by developing an AI-powered exam proctoring system that leverages computer vision and machine learning technologies to detect suspicious behaviors during examinations. The system utilizes strategically placed cameras in exam halls to continuously monitor student activities throughout the examination period. Through advanced behavioral analysis algorithms, the system can automatically identify and flag potentially dishonest actions such as unauthorized eye movements, physical contact between students, whispering or verbal communication, unauthorized material usage, and other suspicious behaviors that may indicate cheating attempts. When the AI detects such activities, it immediately alerts the exam supervisor through a real-time notification system, enabling prompt intervention while maintaining a comprehensive log of all flagged incidents for post-examination review. The primary objectives of this project are threefold: first, to significantly enhance exam security and integrity through continuous, consistent monitoring; second, to reduce dependence on human supervision by automating the detection of suspicious behaviors; and third, to provide an objective, data-driven approach to exam supervision that eliminates subjective bias and ensures fair treatment of all students. The system will deliver several key outputs including real-time suspicious behavior detection with immediate alerts, automated notification delivery to exam supervisors, comprehensive behavioral analysis reports with timestamps and evidence, and a complete video recording archive with searchable incident markers.

for post-examination investigation and dispute resolution. This innovative solution represents a significant advancement in educational technology, offering institutions a powerful tool to maintain academic standards while optimizing resource allocation and ensuring consistent, fair examination conditions for all students.